

DEPARTMENT OF MATHEMATICS

FEDERAL UNIVERSITY, LAFIA

FIRST SEMESTER 2017/2018 EXAMINATION

Course Code: MTH 111 General Mathematics I (3 Units)

Instructions: Answer All Questions

VALEDICTORY

Time: 2:30min

- How many words can be formed from the letters of VALEDICTORY, provided the letters A, E, I, O, Y are not to be separated at all? (A) 7200 (B) 39916800 (C) 3991670 (D) 120 (E) 604800
- The equivalence of set A and B is denoted by (A) $A=B$ (B) $A \Rightarrow B$ (C) $A \rightarrow B$ (D) $A \sim B$ (E) $A \subset B$
- If $n_{p_1} = 12$ n_{p_2} find n. (A) 8 (B) -1 (C) 6 (D) 6 or -1 (E) 8 or -1
- Find the coefficient of x^2 in the expansion of $(1-3x)^{-\frac{5}{2}}$ (A) $\frac{-15}{2}$ (B) $\frac{315}{8}$ (C) $\frac{-316}{8}$ (D) $\frac{15}{2}$ (E) $\frac{315}{4}$
- Evaluate the sum of the first n terms of the arithmetic series $3 + 7 + 11 + 15 + \dots$
(A) $n(2n+1)$ (B) $n(2n-1)$ (C) $\frac{n}{2}(2n+1)$ (D) $4n-1$ (E) $4n+1$
- If the roots of the equation $2x^2 - 3x + r = 0$ are equal, find r. (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $\frac{9}{8}$ (E) $\frac{1}{4}$
- $-2 + 2i$ in polar form is given by (A) $(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$ (B) $\sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$
(C) $2\sqrt{2}(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4})$ (D) $2\sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$ (E) $(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4})$
- Compute $\frac{2+9i}{5-2i}$. (A) $\frac{-8}{29} - \frac{49i}{29}$ (B) $\frac{-8}{29}$ (C) $\frac{49i}{29}$ (D) $\frac{-8}{29} + \frac{49i}{29}$ (E) $\frac{-49i}{29}$
- If $n_{C_6} = n_{C_4}$, find n (A) 7 (B) 10 and 8 (C) 8 (D) 10 (E) -1 and 10
- From question (9) above, find n_{C_2} (A) 28 (B) 45 (C) 21 (D) 73 (E) 88
- Given that $\tan \theta = 1$ and $-360^\circ < \theta < 360^\circ$, what are the four possible values of θ ?
 $\theta = 45^\circ$
 $360 - 45 = 315$
(A) $315^\circ, 135^\circ, -45^\circ, -225^\circ$ (B) $-315^\circ, -135^\circ, 45^\circ, 225^\circ$ (C) $315^\circ, -135^\circ, -45^\circ, 225^\circ$
(D) $-315^\circ, 135^\circ, 45^\circ, -225^\circ$ (E) $-315^\circ, -135^\circ, -45^\circ, -225^\circ$
- Given that $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, compute $A^T A$. (A) $\begin{pmatrix} 25 & 2 \\ 2 & 5 \end{pmatrix}$ (B) $\begin{pmatrix} 25 & 2 \\ -2 & 5 \end{pmatrix}$ (C) $\begin{pmatrix} -25 & 2 \\ -2 & 5 \end{pmatrix}$ (D) $\begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix}$
(E) $\begin{pmatrix} 10 & 5 \\ -2 & 5 \end{pmatrix}$
- If $\theta = \frac{5\pi}{6}$ in radians, what is θ in degrees? (A) 30° (B) 150° (C) 50° (D) 105° (E) 60°
- The modulus of $\frac{7-i}{3-4i}$ is (A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) $\sqrt{3}$ (D) 2 (E) -2
- Simplify $(1 - \cos A)(1 + \sec A)$ (A) $\sin A \tan A$ (B) $\sin A$ (C) $\cos A$ (D) $\sin A \cos A$ (E) $\tan A$

- (B) $x = 2$ or $\frac{-2}{9}$ and $y = -3$ or $\frac{-37}{9}$ (C) $x = 2$ or $\frac{2}{9}$ and $y = 3$ or $\frac{37}{9}$
 (D) $x = -3$ or $\frac{-37}{9}$ and $y = -3$ or $\frac{-37}{9}$ (E) $x = -2$ or 3 and $y = \frac{-2}{9}$ or $\frac{-37}{9}$

17. What is the value of $\det(A)$ if $A = \begin{pmatrix} 2 & 7 & 5 \\ 4 & 6 & 3 \\ 8 & 9 & 1 \end{pmatrix}$? (A) 38 (B) 140 (C) -60 (D) -42 (E) -38

18. The real and the imaginary parts of $4 - i$ are (A) 4 and 1 (B) -1 and 4 (C) 4 and -1 (D) 1 and 4 (E) 4 and 0

19. Simplify $(\cos 2\theta + i \sin 2\theta)^2 (\cos \theta - i \sin \theta)^3$ using DeMoivre's theorem. (A) $(\cos 2\theta + i \sin 2\theta)$ (B) $(\cos 2\theta - i \sin 2\theta)$ (C) $(\cos \theta + i \sin \theta)$ (D) $(\cos 2\theta - i \sin 2\theta)$ (E) $\cos 2\theta$

20. If $B = \begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix}$, what is the value of $B^2 - 4B + 3I$? (A) $\begin{pmatrix} 5 & 1 \\ 3 & -2 \end{pmatrix}$ (B) $\begin{pmatrix} 11 & 1 \\ 3 & 8 \end{pmatrix}$ (C) $\begin{pmatrix} 11 & -1 \\ -3 & 18 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (E) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

21. The descriptive notation for union of two sets A and B is (A) $A \cup B = \{x/x \in A \text{ and } x \in B\}$ (B) $A \cup B = \{x: x \in A \cup x \in B\}$ (C) $A \cup B = \{x/x \in A \text{ or } x \in A\}$ (D) $A \cup B = \{x: x \in A \cap x \in B\}$ (E) $A \cup B = \{x: x \in A < x \leq B\}$

22. In how many ways can 3 boys and 5 girls be arranged in a row so that all the three boys are together. (A) 4320 (B) 40320 (C) 4838400 (D) 2160 (E) 720.

23. If the 21st and 22nd terms in the expansion of $(1+x)^{44}$ are equal. Find the value of x

- (A) 6 (B) 10 (C) $\frac{3}{2}$ (D) $\frac{5}{2}$ (E) $\frac{7}{8}$

24. In a geometric progression, the sum of the 2nd and 3rd term is 9 and the 7th term is eight times the 4th term. Find the common ratio. (A) -3 (B) $\frac{2}{3}$ (C) 3 (D) $\frac{3}{2}$ (E) 2

25. Find the range of values of p if the equation $px^2 - 4(p-1)x + 9 = 0$ has real roots (A) $p \leq 4$ or $p \geq 2$ (B) $p \leq \frac{1}{2}$ or $p \geq 2$ (C) $p \leq \frac{1}{2}$ or $p \geq 3$ (D) $p \leq -\frac{1}{2}$ or $p \geq 4$ (E) $p \leq \frac{1}{4}$ or $p \geq 4$

26. Given that $\begin{pmatrix} 3x & 5 \\ -1 & y \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 8 & -10 \end{pmatrix}$, what is the value of $3x + 2y^2$? (A) -10 (B) 29 (C) -209 (D) -29 (E) 209

27. If $\sin x = \frac{1}{2}$ and x is obtuse, what is $\cot x$? (A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) 3 (E) $-\sqrt{3}$

28. Without solving the quadratic equation, define the nature of the roots of $4x^2 + x - 3 = 0$ (A) real and distinct (B) equal roots (C) not real roots (D) $-1, \frac{3}{4}$ (E) $2, \frac{3}{4}$

29. The compound angle identity $\cos(A+B)$ is given by (A) $\sin A \sin B + \cos A \cos B$ (B) $\cos A \cos B - \sin A \sin B$ (C) $\sin A \sin B - \cos A \cos B$ (D) $\sin A \cos B + \sin B \cos A$ (E) $\cos A \cos B + \sin A \sin B$

30. If $\cos \theta = \frac{\sqrt{3}}{2}$ and $\tan \theta$ is negative, what is $\sin \theta$? (A) $\sin \theta = \frac{1}{2}$ (B) $\sin \theta = -\frac{\sqrt{3}}{2}$ (C) $\sin \theta = -\frac{1}{2}$ (D) $\sin \theta = \frac{3}{2}$ (E) $\sin \theta = \frac{2}{3}$

31. The 2nd term of a geometry series is 24, the 5th term is 81. Find the first term. (A) 14 (B) $\frac{3}{2}$ (C) 12 (D) 16 (E) -12

TIME ALLOWED: $2\frac{1}{2}$ Hours

CREDIT UNIT(S): 3

1. If $z = 3 + 5i$, find $z - \bar{z}$. (A) $10i$ (B) 25 (C) 34 (D) $-4 + 15i$ (E) 6
2. Express $(3 - 8i)(5 + 7i)$ in the $x + iy$ form. (A) $71 + 24i$ (B) $71 - 24i$ (C) $-41 + 66i$ (D) $-71 + 66i$ (E) $-41 - 24i$
3. A matrix A is skew symmetric if (A) $A^T = -A^T$ (B) $A = A^T$ (C) $A' = -A$ (D) $A = -A^T$ (E) $A = -A$
4. Find two numbers such that their arithmetic mean is 6.5 and their geometric mean is 6. (A) $3, 6$ (B) $4, 16$ (C) $6, 8$ (D) $4, 9$ (E) $5, 7$
5. Find the modulus of $\cos \theta + i \sin \theta$. (A) $-i$ (B) -1 (C) $2i$ (D) i (E) 1
6. If $\theta = \frac{7\pi}{6}$ in radians, what is θ in degrees? (A) 45° (B) 125° (C) 205° (D) 220° (E) 210°
7. Calculate the argument of $\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi$ (A) π (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$ (E) $\frac{3\pi}{2}$
8. The value of $\tan(-15^\circ)$ in surd form is..... (A) $2 - \sqrt{3}$ (B) $\sqrt{3} + 2$ (C) $\sqrt{3} - 2$ (D) $2 + \sqrt{3}$ (E) $\sqrt{3} + 3$
9. Given that $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, find $A^T \cdot A$. (A) $\begin{pmatrix} 21 & 2 \\ -2 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 25 & -2 \\ -2 & 3 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 20 & -2 \\ -4 & 1 \end{pmatrix}$ (E) $\begin{pmatrix} 4 & 2 \\ 0 & -5 \end{pmatrix}$
10. If $\cos \theta = \frac{\sqrt{3}}{2}$ and $\tan \theta$ is negative, what is $\sin \theta$? (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $-\frac{\sqrt{3}}{2}$ (E) $\frac{\sqrt{2}}{2}$
11. Find the number of ways of selecting 3 prefects from a class of 18 students. (A) 816 (B) 4896 (C) 18 (D) 1253 (E) 81
12. Three consecutive terms in a geometric sequence are $n - 2, n$ and $n + 3$. What is the common ratio of the sequence? (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{3}{2}$ (E) $\frac{4}{3}$
13. A diagonal matrix whose diagonal elements are one (1) is called (A) Triangular matrix (B) Diagonal Matrix (C) Unit Matrix (D) Null Matrix (E) Square Matrix
14. The cardinality of the power set of A where $A = \{b, c, d, e\}$ is (A) 4 (B) 10 (C) 8 (D) 16 (E) 32
15. In a class of 40 students, 24 play football, 18 play hockey and 6 do not play any game. How many play both? (A) 3 (B) 2 (C) 5 (D) 1 (E) 8
16. The complex number $\sqrt{3} - i$ has modulus as (A) -3 (B) 4 (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$ (E) 2
17. Find the coefficient of x^4 in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$. (A) 15 (B) ${}^{15}C_3$ (C) ${}^{12}C_5$ (D) ${}^{12}C_8$ (E) 120
18. If $\frac{2+9i}{5-2i} = x + iy$, where x and y are real. What are the values of x and y ? (A) $\left(-\frac{9}{2}, \frac{2}{5}\right)$ (B) $\left(\frac{2}{5}, -\frac{9}{2}\right)$ (C) $\left(-\frac{8}{29}, \frac{49}{29}\right)$ (D) $\left(\frac{49}{29}, \frac{8}{29}\right)$ (E) $\left(\frac{49}{29}, \frac{8}{29}\right)$
19. If $A = \{1, 2, a, b, c, d\}$, $B = \{2, 3, a, b, c, d, e, f\}$, $C = \{2, 4, c, d, h\}$ and $D = \{1, a, 2, b, 3, c\}$. Obtain $(A \cap B) \cap (C \cup D)$ (A) $\{1, b\}$ (B) $\{2, a, b, c, d\}$ (C) $\{2, 1, a, b, c\}$ (D) $\{b, 4, h\}$ (E) $\{1, a, b, 3\}$

$$n^2 - n^2 + 2n - 3 = 0$$

$$n-1, n, n+3$$

$$\frac{n}{n-1} \neq \frac{n+3}{n}$$

$$n^2 - n^2 + 3n - 3 = 0$$

X

20. For what values of x is $2 \cos^2 x - \sin x = 1$ for $x \in [0, 2\pi]$.
 (A) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (B) $\frac{\pi}{6}, \frac{2\pi}{3}, \frac{3\pi}{4}$
 (C) $0, \frac{\pi}{3}, \frac{2\pi}{3}$ (D) $0, \pi, \frac{3\pi}{2}$ (E) $\frac{\pi}{3}, \frac{2\pi}{3}, \pi$
21. Given that $\begin{pmatrix} 3x & 5 \\ -1 & y \end{pmatrix} = \begin{pmatrix} 9 & 5 \\ -1 & -10 \end{pmatrix}$, Find the values of $3x - 2y$?
 (A) 29 (B) -28 (C) 13 (D) 10 (E) -27
22. Solve the equation $\cos x + \sin x = \frac{1}{\cos x - \sin x}$ for values of x such that $0 \leq x < 2\pi$.
 (A) $\frac{\pi}{2}, \frac{3\pi}{2}$ (B) $0, \pi$ (C) $0, \frac{\pi}{3}$ (D) $\frac{\pi}{3}, \pi$ (E) $\frac{\pi}{3}, \frac{2\pi}{3}$
23. The determinant of $P = \begin{pmatrix} \cos 30^\circ & \sin 30^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix}$ is (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 0 (D) $\frac{\sqrt{3}}{4}$ (E) $\frac{3}{2}$
24. If $\cos \theta = \frac{12}{13}$, what is the value of $\sin \theta$? (A) $\frac{13}{5}$ (B) $\frac{6}{13}$ (C) $\frac{13}{6}$ (D) $\frac{5}{13}$ (E) $\frac{5}{12}$
25. The real and imaginary parts of $\frac{7-i}{1+7i}$ are respectively. (A) $\frac{7}{25}, -1$ (B) $\frac{14}{50}, -\frac{48}{50}$ (C) $\frac{9}{5}, -\frac{4}{5}$
 (D) $\frac{4}{5}, \frac{9}{5}$ (E) $0, -1$
26. Given two matrices S and T with orders (4×5) and (5×6) respectively. The matrix ST has order
 (A) (5×5) (B) (4×5) (C) (5×4) (D) (5×6) (E) (4×6)
27. There are basically _____ step(s) in the proof by mathematical induction. (A) 1 (B) 4 (C) 3 (D) 2 (E) 5
28. The minor of the entry, a_{23} of a given matrix is -14. What is the value of the cofactor of the entry?
 (A) -14 (B) 13 (C) 14 (D) 15 (E) -13
29. If the sum of infinity of the series $x + x^3 + x^5 + \dots$ is $\frac{2}{3}$, find the value(s) of x . (A) $\frac{1}{2}$ (B) $\frac{1}{2}, -2$
 (C) $\frac{1}{2}, 2$ (D) 2 (E) 3
30. If $R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, what is R^{-1} ? (A) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (C) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (D) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ (E) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$
31. Solve within the range $0^\circ \leq x \leq 180^\circ$ the equation, $\sin x + 2 \sin x \cos x = 0$. (A) $x = 0^\circ, 120^\circ, 180^\circ$
 (B) $x = 0^\circ, 150^\circ$ (C) $x = 180^\circ, 360^\circ$ (D) $x = 0^\circ, 150^\circ, 180^\circ, 360^\circ$ (E) $x = 90^\circ, 180^\circ$
32. Given $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, evaluate $f(A) = A^2 - 4A + 3I$. (A) $\begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
 (C) $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$ (D) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (E) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
33. Find the determinant of the matrix, $A = \begin{pmatrix} 2 & -6 & 10 \\ 4 & 0 & -2 \\ 3 & -2 & 8 \end{pmatrix}$.
 (A) 0 (B) 208 (C) -316 (D) -26 (E) 120
34. Using the first two terms of series in the binomial expansion of $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} + \dots$.
 Evaluate $\sqrt{3}$. (A) $-\frac{1}{2}$ (B) $\frac{1}{4}$ (C) -2 (D) $\frac{7}{4}$ (E) $\frac{1}{2}$
35. What is the n th term of the sequence 9, 3, 1, ...?
 (A) $9 \left(\frac{1}{3}\right)^{n+1}$ (B) $3 \left(\frac{1}{3}\right)^{n-1}$ (C) $27 \left(\frac{1}{3}\right)^{n+1}$ (D) $9 \left(\frac{1}{3}\right)^n$ (E) $27 \left(\frac{1}{3}\right)^n$
36. If the 9th term of an arithmetic sequence is 5 times the 5th term, find the relationship between the first term, a and the common difference, d .
 (A) $a + 2d = 0$ (B) $2a + 3d = 0$ (C) $3a + d = 0$ (D) $2a + d = 0$ (E) $a + 3d = 0$